Supporting information for: Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

This supporting information accompanies the manuscript titled *Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling* and provides expanded workflow notes and supplemental derivations.

S1 Bandwidth scaling for the specular Bragg-sphere cap overlap

This section gives the short reciprocal-space derivation behind the statement that a fixed Ewald-shell thickness samples a larger angular fraction of a low-order $00L$ Bragg sphere than of a high-order one.

For the specular family, $m = 0$ and $G_R = 0$, so

$$G_L \equiv G_{00L} = \frac{2\pi L}{c}. \tag{S1}$$

Thus the Bragg-sphere radius grows linearly with L.

In the fixed-center Ewald-shell construction used for the geometric interpretation, the central wavevector is

$$K_0 = \frac{2\pi}{\lambda_0}. \tag{S2}$$

A small fractional bandwidth $b = \Delta\lambda/\lambda_0$ gives a reciprocal-space shell thickness

$$\Delta K \simeq K_0 b. \tag{S3}$$

This thickness is set by the instrument and is independent of L.

Let ψ be the angular position of the Ewald-shell intersection band on the Bragg sphere. For an Ewald radius K, with the shell center held fixed at the central K_0 position, the intersection satisfies

$$\sin\psi(K; G_L) = \frac{K_0^2 - K^2 + G_L^2}{2K_0 G_L}. \tag{S4}$$

At the central wavelength, $K = K_0$, this reduces to

$$\sin\psi_0 = \frac{G_L}{2K_0}. \tag{S5}$$

Differentiating the fixed-center relation at $K = K_0$ gives

$$\cos\psi_0 \, d\psi \simeq -\frac{dK}{G_L}. \tag{S6}$$

Therefore the full angular width of the Ewald intersection band is approximately

$$\Delta\psi \simeq \frac{\Delta K}{G_L \cos\psi_0} \simeq \frac{K_0 b}{G_L \cos\psi_0}. \tag{S7}$$

Substituting $G_L = 2\pi L/c$ and $K_0 = 2\pi/\lambda_0$ gives

$$\Delta\psi(L) \simeq \frac{bc}{\lambda_0 L \cos\psi_0}. \tag{S8}$$

For low and moderate L, $\cos\psi_0$ varies more slowly than G_L, so the leading scaling is

$$\Delta\psi(L) \propto \frac{1}{L}. \tag{S9}$$

The mosaic cap clips this band. If the cap has angular half-width η and is displaced by the nominal incident angle θ_i, a one-dimensional angular measure of the accessible overlap is

$$\ell(L) = \max\left[0, \min\left(\psi_0 + \frac{\Delta\psi}{2}, \theta_i + \eta\right) - \max\left(\psi_0 - \frac{\Delta\psi}{2}, \theta_i - \eta\right)\right]. \tag{S10}$$

When the Ewald band is narrower than the mosaic cap and limits the overlap, $\ell(L) \simeq \Delta\psi(L)$, so the bandwidth-controlled accessible overlap inherits the same approximate $1/L$ scaling. If the cap is fully inside the band, the overlap is mosaic-limited instead. If the band and cap do not intersect, the overlap is zero.

This scaling describes the fixed-center reciprocal-space shell overlap used to interpret the Bragg-sphere schematics. It is not the raw Bragg-sphere surface area, and it is distinct from the strict moving-center Bragg-law derivative $\Delta\theta_B \simeq b \tan\theta_B$.

S2 Staged refinement workflow (expanded)

1. **Beam characterization (direct beam).** Acquire direct-beam images at several detector distances. Fit the spot position and shape versus distance to determine the incident-beam widths (w_x, w_z) and Gaussian divergences $(\Delta\theta_x, \Delta\theta_z)$ that set the instrument-limited broadening.

2. **Detector geometry calibration (3D powder standard).** Measure a polycrystalline hBN calibrant and subtract the detector dark frame. Fit multiple Debye–Scherrer rings in detector coordinates and refine detector distance, detector tilts (β, γ), in-plane detector rotation χ, and beam center (x_0, y_0).

3. **Sample alignment and goniometer misalignment.** For each dark-subtracted sample image, use indexed Bragg-peak centers rather than full-image intensity. With detector parameters fixed, refine sample alignment by minimizing detector-space residuals between measured and calculated peak positions spanning both specular and off-specular regions. Refine $(\theta_i, \delta, z_B, z_S)$ and the goniometer-axis misalignment rotations (α, ψ); include film dimensions (W, H) and thickness t to weight footprint and absorption.

4. **Mosaicity/profile refinement.** With geometry fixed, compare selected local detector ROIs or local $I(\phi)$ profiles after center alignment. The shared profile model is a wrapped Gaussian-plus-Lorentzian angular distribution, not a pseudo-Voigt common-width peak

approximation. Off-specular arcs mainly constrain the Gaussian core, while off-condition specular intensity constrains the Lorentzian tail. Report Lorentzian FWHM, Gaussian FWHM, and the relative Lorentzian weight.

5. **Ordered-structure refinement.** With geometry and mosaicity fixed, refine the ordered intensity model against background-subtracted integrated counts in selected detector-space Bragg regions of interest (ROIs) from multiple dark-subtracted images. The corresponding prediction is the calculated detector intensity deposited in those same ROIs. Use one nuisance scale per image and physically constrained ordered-structure parameters.

6. **Disorder refinement.** After the ordered model converges, introduce stacking-disorder parameters only through fixed-Q_r rod profiles reported versus Q_z. The measured profile is formed from caked signal and normalization fields by summing signal and normalization over the selected rod mask before division. Geometry, detector calibration, beam terms, mosaicity, lattice constants, and the ordered baseline remain fixed; only stacking-disorder parameters and declared scale/background terms are refined.

All detector images are dark-subtracted before fitting. Any additional preprocessing (flat-field, polarization, solid-angle, or detector-efficiency corrections) should either be applied consistently to all images before refinement or stated explicitly if omitted. We recommend stage-specific uncertainty estimation by repeated constrained refits: resample the calibrant points for detector geometry, indexed peak lists for sample alignment, local reflection regions for mosaicity, accepted Bragg ROIs for ordered intensity, and selected rod bins for disorder. Report the standard deviation of the accepted solutions and propagate parameters to the next stage only after the preceding stage is stable.

S2.1 Coordinate remapping, projected profiles, and uncertainty propagation

The detector image remains the primary data object. The 2θ–ϕ representation is an overlap-weighted remapping used to evaluate fixed-m or fixed-Q_r trajectory masks and projected Q_z or L profiles. It is not treated as an independent measurement. The same remapping, masks, and profile extraction are applied to measured and calculated detector images.

The caked image is constructed with a sparse detector-to-cake operator

$$M \in \mathbb{R}^{B\times P}, \tag{S11}$$

where P is the number of detector pixels, B is the number of caked bins, and M_{bk} is the fractional overlap weight from detector pixel k into caked bin b. For detector signal s_k and normalization n_k,

$$S_b = \sum_k M_{bk} s_k, \qquad N_b = \sum_k M_{bk} n_k, \qquad I_b = \frac{S_b}{N_b}, \tag{S12}$$

for bins with $N_b > 0$. Signal and normalization are therefore accumulated before division. This prevents a projected profile from becoming a sum of already normalized pixels and keeps the observable tied to the detector-space calibration.

For each remapped bin, the calibrated detector and sample geometry define a scattering vector $\mathbf{Q}$. The coordinates used to label selected trajectories are

$$Q_r = |\mathbf{Q}_{\parallel}|, \qquad Q_z = \mathbf{Q}\cdot\hat{\mathbf{n}}, \qquad L = \frac{Q_z}{|\mathbf{c}^*|} = \frac{Q_z c}{2\pi}, \tag{S13}$$

where $\hat{\mathbf{n}}$ is the film normal. A selected fixed-Q_r or fixed-m family is retained when

$$|Q_r - Q_{r,0}| \le \Delta Q_r, \qquad Q_{z,\min} \le Q_z \le Q_{z,\max}. \tag{S14}$$

This region is simple in the physical Q_r–Q_z description but appears curved in the displayed 2θ–ϕ map because the coordinate transformation is nonlinear. The curved shape is therefore a consequence of the projection, not an additional physical model.

For a selected profile coordinate ℓ, let $r_{\ell b}$ be the mask or interpolation weight that assigns caked bin b to that profile point. The profile is

$$I_\ell^{\mathrm{ROI}} = \frac{\sum_b r_{\ell b} S_b}{\sum_b r_{\ell b} N_b} = \frac{\sum_k a_{\ell k} s_k}{\sum_k a_{\ell k} n_k}, \qquad a_{\ell k} = \sum_b r_{\ell b} M_{bk}. \tag{S15}$$

Each point in the plotted profile is therefore an average over a finite detector-space band. Its coordinate may be labeled by Q_z or L, but the point represents a resolution-limited projection rather than the intensity at one exact reciprocal-space coordinate.

The transformed-coordinate uncertainty has two parts: uncertainty in the coordinate label and uncertainty in the rebinned intensity. The coordinate-label uncertainty follows from the finite detector-pixel footprint and from uncertainty in the detector calibration. For square detector pixels with pitch p, detector distance d, and $t = 2\theta$, the leading flat-detector terms are

$$\sigma_{2\theta,\mathrm{pix}}(t) = \frac{p\cos^2 t}{\sqrt{12}\,d}, \qquad \sigma_{\phi,\mathrm{pix}}(t) = \frac{p}{\sqrt{12}\,d\tan t}. \tag{S16}$$

The radial term is finite and largest near the direct beam. The azimuthal term scales as $1/\tan(2\theta)$, so the problematic regime is confined to the near-specular, low-2θ region where azimuth is poorly conditioned.

For the $d = 75$ mm and $p = 300\,\mu$m geometry used in the uncertainty calculation, the maximum radial coordinate uncertainty is approximately 0.066°. The azimuthal coordinate uncertainty falls below 1° above $2\theta \approx 3.8^\circ$, below 0.5° above $2\theta \approx 7.5^\circ$, and below 0.25° above $2\theta \approx 14.9^\circ$. More generally, the threshold for a target azimuthal width w_ϕ is

$$2\theta_{\mathrm{crit}} = \arctan\left(\frac{p}{\sqrt{12}\,d\,w_\phi}\right), \tag{S17}$$

with w_ϕ expressed in radians.

Additional calibration uncertainty can be propagated with the usual Jacobian form. If the uncertain geometry parameters are collected in $\mathbf{g}$ with covariance matrix Σ_g, then

$$\sigma^2_{2\theta,\mathrm{cal}} = J_{2\theta}\Sigma_g J_{2\theta}^{\mathsf{T}}, \qquad \sigma^2_{\phi,\mathrm{cal}} = J_\phi \Sigma_g J_\phi^{\mathsf{T}}. \tag{S18}$$

If the displayed bin-center label is also treated as uncertain, the finite bin width contributes

$$\sigma^2_{2\theta,\mathrm{bin}} = \frac{\Delta(2\theta)^2}{12}, \qquad \sigma^2_{\phi,\mathrm{bin}} = \frac{\Delta\phi^2}{12}. \tag{S19}$$

These coordinate terms describe the uncertainty of the transformed axis value. They are separate from intensity uncertainty.

The intensity uncertainty is inherited from the detector through the same overlap weights used to construct the caked image and the projected profile. If v_k is the detector-pixel variance and the detector-pixel variances are independent, then

$$\mathrm{Var}\Big(I_\ell^{\mathrm{ROI}}\Big) = \frac{\sum_k a_{\ell k}^2 v_k}{\left(\sum_k a_{\ell k} n_k\right)^2}. \tag{S20}$$

If overlapping interpolation weights are used, neighboring profile points can be statistically correlated. The corresponding covariance is

$$\mathrm{Cov}\Big(I_\ell^{\mathrm{ROI}}, I_q^{\mathrm{ROI}}\Big) = \frac{\sum_k a_{\ell k} a_{qk} v_k}{\left(\sum_k a_{\ell k} n_k\right)\left(\sum_k a_{qk} n_k\right)}. \tag{S21}$$

Thus the same weights that define the projected profile also propagate the detector-pixel variance. The covariance between neighboring projected points is neglected in the present line-shape comparison unless formal profile error bars are required.

The displayed coordinate for a projected point can be represented by the same finite-band weighting. For example,

$$\bar{L}_\ell = \frac{\sum_b r_{\ell b} N_b L_b}{\sum_b r_{\ell b} N_b}, \qquad \sigma_{L,\ell}^2 = \frac{\sum_b r_{\ell b} N_b (L_b - \bar{L}_\ell)^2}{\sum_b r_{\ell b} N_b}, \tag{S22}$$

before adding calibration, incidence-angle, wavelength, divergence, and mosaic contributions. This width is a projection and resolution effect. It is not a counting-statistical error bar.

These projection uncertainties are coupled to model-parameter correlations. Peak positions are sensitive to detector distance, beam center, detector tilt, sample height, and incidence angle. Profile widths are sensitive to both beam divergence and the Gaussian mosaic core. Weak off-condition and low-L intensities are sensitive to wavelength bandwidth, background, and the Lorentzian mosaic tail. Relative integrated intensities are sensitive to scale, footprint, absorption, thickness, and structure-factor parameters. The staged workflow separates these effects by fixing or constraining the parameters tied to peak position before fitting profile widths, and by fixing the geometry and mosaic state before interpreting ordered intensities.

The practical consequences are: (i) the 2θ–ϕ and Q_r/Q_z profiles are controlled projections of the detector image, not independent ideal reciprocal-space scans; (ii) transformed-coordinate uncertainty is concentrated near the low-angle/specular region; (iii) intensity variance is propagated through the same overlap weights used for caking and profile extraction; and (iv) the strongest remaining systematic coupling occurs near low L, where direct-beam background, bandwidth, divergence, incidence angle, and the Lorentzian mosaic tail contribute to the same detector-space feature.

S2.2 Wrapped mosaic distribution

In the main text we describe out-of-plane mosaicity by a misorientation density $\omega(\Delta\theta)$ in the crystallite tilt angle $\Delta\theta$ relative to the substrate normal. Since $\Delta\theta$ is an orientation angle, $\Delta\theta$ and $\Delta\theta + 2\pi$ represent the same physical state. Accordingly, ω is treated as a 2π-periodic probability density on $\Delta\theta \in (-\pi, \pi]$.

The implemented profile is a wrapped Gaussian plus a wrapped Lorentzian,

$$\omega_{\mathrm{w}}(\Delta\theta; \eta, \Gamma, \sigma) = \eta\, L_{\mathrm{w}}(\Delta\theta; \Gamma) + (1-\eta)\, G_{\mathrm{w}}(\Delta\theta; \sigma), \qquad 0 \le \eta \le 1.$$

This is not a pseudo-Voigt constraint: the Lorentzian half-width Γ and Gaussian standard deviation σ are independent. The wrapped Lorentzian and wrapped Gaussian are

$$L_{\mathrm{w}}(\Delta\theta;\Gamma) = \sum_{m=-\infty}^{\infty} \frac{1}{\pi} \frac{\Gamma}{(\Delta\theta + 2\pi m)^2 + \Gamma^2} = \frac{1}{2\pi} \frac{\sinh\Gamma}{\cosh\Gamma - \cos\Delta\theta},$$

$$\begin{aligned} G_{\mathrm{w}}(\Delta\theta;\sigma) &= \sum_{m=-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(\Delta\theta + 2\pi m)^2}{2\sigma^2}\right] \\ &= \frac{1}{2\pi}\left[1 + 2\sum_{n=1}^{\infty} e^{-n^2\sigma^2/2} \cos(n\Delta\theta)\right]. \end{aligned}$$

Both components integrate to unity over $(-\pi, \pi]$, so ω_{w} is normalized. The reported Lorentzian FWHM is 2Γ, and the reported Gaussian FWHM is $2\sqrt{2\ln 2}\,\sigma$. The local small-width limit recovers the usual line-shape widths, but the refinement and normalization are defined by the wrapped functions above.

For a reflection with Bragg-sphere radius $G \equiv |\mathbf{G}|$, mosaicity redistributes intensity in polar angle ϑ with line density $I_0\,\omega_{\mathrm{w}}(\vartheta - \vartheta_B)\,d\vartheta$, where ϑ_B is the Bragg position for that reflection. Random in-plane rotations about the film normal revolve this line density about the G_z axis, yielding an axially symmetric surface density $\rho(\vartheta)$. With area element $dA = 2\pi G^2 \sin\vartheta\,d\vartheta$, conservation of intensity gives

$$\rho(\vartheta) = \frac{I_0}{2\pi G^2 \sin\vartheta}\,\omega_{\mathrm{w}}(\vartheta - \vartheta_B), \qquad 0 < \vartheta < \pi,$$

as used in the main text.

S2.3 Wave propagation, refraction, and absorption

Wave attenuation occurs both at the interfaces and as the wave propagates through the film. In the forward model we therefore (i) evaluate the complex refractive index for each sampled wavelength in the source bandwidth, (ii) refract each ray into and out of the film, (iii) determine the complex surface-normal component of the internal wavevector from the boundary condition at the interface, and (iv) weight scattering from a finite thickness t by the corresponding transmission and propagation attenuation. This subsection records the explicit equations used.

Complex refractive index over the source bandwidth. For a wavelength sample λ_j with spectral weight w_j, the vacuum wavevector magnitude is

$$k_0(\lambda_j) = \frac{2\pi}{\lambda_j}. \tag{S23}$$

The material optical response is represented by the wavelength-dependent complex refractive index

$$\tilde{n}(\lambda_j) = 1 - \delta_n(\lambda_j) + i\beta_n(\lambda_j), \tag{S24}$$

where the subscript n avoids confusion with geometric angles used elsewhere. The real correction δ_n is the refractive decrement that shifts the phase velocity and bends the ray inside the film. The imaginary correction β_n gives absorption and sets the finite penetration depth. The real decrement may be estimated from electron density ρ_e and classical electron radius r_e, while the imaginary correction is the wavelength-dependent optical input used directly by the complex-wavevector calculation:

$$\delta_n(\lambda_j) = \frac{r_e \lambda_j^2 \rho_e}{2\pi}, \qquad \beta_n(\lambda_j) = \mathrm{Im}\{\tilde{n}(\lambda_j)\}. \tag{S25}$$

The bandwidth therefore changes both the Ewald-sphere radius through $k_0(\lambda_j)$ and the optical constants that determine refraction, transmission, and attenuation. The attenuation factor below is not applied through a separate scalar absorption coefficient; it is computed from the imaginary parts of the refracted internal wavevectors.

Angles and refraction. Let $\hat{z}$ be the sample normal. For an incident ray in air with wavevector $\mathbf{k}_i(\lambda_j)$, the grazing-incidence angle θ_i and in-plane azimuth φ_i are

$$\sin\theta_i(\lambda_j) = \frac{k_{i,z}(\lambda_j)}{k_0(\lambda_j)}, \tag{S26}$$

$$\tan\varphi_i = \frac{k_{i,x}(\lambda_j)}{k_{i,y}(\lambda_j)}. \tag{S27}$$

Upon refraction into the film, the transmitted grazing angle is written in the same form used in the dissertation wave-propagation construction,

$$\theta_t(\lambda_j) = \cos^{-1}\left[\frac{\cos\theta_i(\lambda_j)}{\mathrm{Re}\{\tilde{n}(\lambda_j)\}}\right]. \tag{S28}$$

The internal wavevector scale used for the in-plane components is

$$k'(\lambda_j) = k_0(\lambda_j)\left[\mathrm{Re}\{\tilde{n}(\lambda_j)^2\}\right]^{1/2}. \tag{S29}$$

The in-plane internal components are therefore

$$k_{t,x}(\lambda_j) = k'(\lambda_j)\cos\theta_t(\lambda_j)\sin\varphi_i, \tag{S30}$$

$$k_{t,y}(\lambda_j) = k'(\lambda_j)\cos\theta_t(\lambda_j)\cos\varphi_i. \tag{S31}$$

The tangential components are fixed at the boundary. The remaining surface-normal component is complex and carries the attenuation of the internal field.

Complex normal component. To isolate the real and imaginary parts of the normal component, write

$$k_{t,z}(\lambda_j)^2 = a(\lambda_j) + ib(\lambda_j), \qquad k_{t,z}(\lambda_j) = u(\lambda_j) + iv(\lambda_j). \tag{S32}$$

Expanding $(u+iv)^2$ gives

$$(u+iv)^2 = u^2 - v^2 + 2iuv = a + ib. \tag{S33}$$

Thus

$$a = u^2 - v^2, \qquad b = 2uv, \qquad |k_{t,z}|^2 = u^2 + v^2 = \sqrt{a^2+b^2}. \tag{S34}$$

Solving for u^2 and v^2 gives

$$u^2 = \frac{\sqrt{a^2+b^2}+a}{2}, \qquad v^2 = \frac{\sqrt{a^2+b^2}-a}{2}. \tag{S35}$$

For the incident internal wavevector,

$$\mathrm{Re}\{k_{t,z}(\lambda_j)^2\} = k'(\lambda_j)^2 \sin^2\theta_t(\lambda_j) \equiv a_t(\lambda_j), \tag{S36}$$

$$\mathrm{Im}\{k_{t,z}(\lambda_j)^2\} = k_0(\lambda_j)^2\,\mathrm{Im}\{\tilde{n}(\lambda_j)^2\} \equiv b(\lambda_j). \tag{S37}$$

The normal component used for propagation inside the film is therefore

$$\mathrm{Re}\{k_{t,z}(\lambda_j)\} = \left[\frac{\sqrt{a_t(\lambda_j)^2 + b(\lambda_j)^2} + a_t(\lambda_j)}{2}\right]^{1/2}, \tag{S38}$$

$$\mathrm{Im}\{k_{t,z}(\lambda_j)\} = \left[\frac{\sqrt{a_t(\lambda_j)^2 + b(\lambda_j)^2} - a_t(\lambda_j)}{2}\right]^{1/2}. \tag{S39}$$

The physical branch is chosen so that $\mathrm{Im}\{k_{t,z}\} \geq 0$, giving a decaying wave inside the film. Equivalently, the same construction can be read as the boundary-condition statement

$$k_{t,z}(\lambda_j)^2 = [k_0(\lambda_j)\tilde{n}(\lambda_j)]^2 - k_{t,x}(\lambda_j)^2 - k_{t,y}(\lambda_j)^2. \tag{S40}$$

This is the implementation form of the $a + ib$ decomposition above.

Exit path and refraction out of the film. The same construction is applied to the outgoing internal wavevector. For the scattered internal angles (θ'_t, φ_f),

$$k'_{t,x}(\lambda_j) = k'(\lambda_j)\cos\theta'_t(\lambda_j)\sin\varphi_f, \tag{S41}$$

$$k'_{t,y}(\lambda_j) = k'(\lambda_j)\cos\theta'_t(\lambda_j)\cos\varphi_f, \tag{S42}$$

$$a_f(\lambda_j) = k'(\lambda_j)^2\sin^2\theta'_t(\lambda_j), \qquad b_f(\lambda_j) = b(\lambda_j). \tag{S43}$$

Hence

$$\mathrm{Re}\{k'_{t,z}(\lambda_j)\} = \left[\frac{\sqrt{a_f(\lambda_j)^2 + b(\lambda_j)^2} + a_f(\lambda_j)}{2}\right]^{1/2}, \tag{S44}$$

$$\mathrm{Im}\{k'_{t,z}(\lambda_j)\} = \left[\frac{\sqrt{a_f(\lambda_j)^2 + b(\lambda_j)^2} - a_f(\lambda_j)}{2}\right]^{1/2}. \tag{S45}$$

Once the internal scattered wavevector is known, the ray is refracted back into air by optical reversibility. In the angle form, inversion of Eq. (S28) gives

$$\theta_f(\lambda_j) = \cos^{-1}\left[\mathrm{Re}\{\tilde{n}(\lambda_j)\}\cos\theta'_t(\lambda_j)\right]. \tag{S46}$$

Fresnel transmission factors. Each ray is weighted by transmission at the entrance and exit interfaces. For the scalar field used in the forward model, the amplitude transmission coefficient between media 1 and 2 is written in terms of normal wavevector components as

$$T_{12} = \frac{2k_{1z}}{k_{1z} + k_{2z}}. \tag{S47}$$

Thus, for air-to-film transmission and film-to-air exit,

$$T_{\rm in}(\lambda_j) = \frac{2k_{i,z}^{(0)}(\lambda_j)}{k_{i,z}^{(0)}(\lambda_j) + k_{t,z}(\lambda_j)}, \qquad T_{\rm out}(\lambda_j) = \frac{2k'_{t,z}(\lambda_j)}{k'_{t,z}(\lambda_j) + k_{f,z}^{(0)}(\lambda_j)}. \tag{S48}$$

The corresponding intensity factor is

$$W_T(\lambda_j) = |T_{\rm in}(\lambda_j)|^2 |T_{\rm out}(\lambda_j)|^2. \tag{S49}$$

Polarization-resolved Fresnel factors can be substituted in the same location if needed. The scalar form is the approximation used for the detector-space weighting here.

Absorption weight from the imaginary normal wavevector. The imaginary parts of the incident and exit normal components determine the propagation damping. For a candidate with entry and exit propagation lengths $L_{\rm in}$ and $L_{\rm out}$, define

$$\kappa_i(\lambda_j) = \mathrm{Im}\{k_{t,z}(\lambda_j)\}, \qquad \kappa_f(\lambda_j) = \mathrm{Im}\{k'_{t,z}(\lambda_j)\}. \tag{S50}$$

The propagation factor is then

$$W_{\rm prop}(\lambda_j) = \exp\left[-2\kappa_i(\lambda_j)L_{\rm in}(\lambda_j) - 2\kappa_f(\lambda_j)L_{\rm out}(\lambda_j)\right]. \tag{S51}$$

This is the absorption/evanescence form used by the simulation. The factor of two converts field amplitude damping into intensity damping. The exponent is dimensionless, so the path lengths must be expressed in the reciprocal units of the wavevectors. If a finite film thickness is supplied, $L_{\rm in}$ and $L_{\rm out}$ are the corresponding slab propagation lengths; otherwise the attenuation is controlled by the penetration depths set by $1/(2\kappa_i)$ and $1/(2\kappa_f)$.

For a uniformly scattering film of thickness t, the same normal-component damping can be written as a depth average,

$$\overline{W}_{\rm prop}(\lambda_j) = \frac{1}{t}\int_0^t \exp[-2\left(\kappa_i(\lambda_j) + \kappa_f(\lambda_j)\right) z]\, dz \tag{S52}$$

$$= \frac{1 - \exp[-2\left(\kappa_i(\lambda_j) + \kappa_f(\lambda_j)\right) t]}{2\left(\kappa_i(\lambda_j) + \kappa_f(\lambda_j)\right) t}. \tag{S53}$$

The total optical weight multiplying the structure-factor and mosaic weights is

$$W_{\rm opt}(\lambda_j) = W_T(\lambda_j) W_{\rm prop}(\lambda_j), \tag{S54}$$

or $W_T\overline{W}_{\rm prop}$ when the depth-averaged form is used. Near the critical-angle region this complex-k_z form is the relevant statement because refraction, evanescence, and finite penetration depth are all carried by the same wavelength-dependent optical constants.

Detector accumulation over the bandwidth. For each accepted scattering event, the detector contribution at wavelength λ_j is weighted by transmission, absorption, structure factor, mosaic probability, beam divergence, footprint, and solid-angle terms before detector deposition. The wavelength-resolved images are then added incoherently,

$$I_{\rm det}(x, y) = \sum_j w_j\, I_{\rm det}[x, y; \lambda_j, \tilde{n}(\lambda_j)]. \tag{S55}$$

This is why finite bandwidth is coupled to refraction rather than only to the Ewald-sphere thickness: changing λ_j changes k_0, $\tilde{n}$, the transmitted angle, the complex normal wavevector, the transmission factors, and the attenuation length.

S2.4 Detector-derived ordered and disorder observables

The detector-space forward model predicts full images, but the inverse problems use stage-specific observables. In the ordered-structure stage, the measured quantity for ROI i in image m is the background-subtracted detector count

$$y_{mi} = \sum_{(u,v)\in\Omega_{mi}} \left[I_m^{\mathrm{meas}}(u,v) - b_{mi}(u,v)\right],$$

where Ω_{mi} is the selected detector-space Bragg ROI and b_{mi} is the local background model. The corresponding prediction is the calculated detector intensity deposited in the same ROI,

$$\mu_{mi} = \sum_{j\in\mathcal{R}_{mi}} w_j,$$

with geometry, beam terms, and mosaicity fixed. One nuisance scale per image accounts for exposure and flux differences. This makes the ordered stage a multi-image ROI-integrated Bragg-intensity refinement, not a full-image pixel objective.

The PbI_2 disorder refinement is a final-stage selected-profile fit, not a second global image refinement. The upstream beam, detector, sample alignment, mosaic/profile parameters, lattice constants, and ordered PbI_2 baseline are fixed before this stage starts. The detector image is first caked into a signal field $S(\phi, 2\theta)$ and a normalization field $N(\phi, 2\theta)$ using the calibrated detector geometry. For the selected-rod conversion, let η denote the polar angle of the scattering vector relative to the film normal. For each caked bin,

$$Q = \frac{4\pi}{\lambda}\sin\left(\frac{2\theta}{2}\right), \qquad Q_r = |\sin\eta| Q, \qquad Q_z = \cos\eta\, Q.$$

A selected rod with in-plane radius $Q_{r,0}$ is retained when

$$|Q_r - Q_{r,0}| \leq \Delta Q_r, \qquad Q_{z,\min} \leq Q_z \leq Q_{z,\max}.$$

A branch selector may be applied after this reciprocal-space mask so that one branch is used for the default fit and the symmetry-related branch is retained for validation.

Let S_{ij} and N_{ij} denote the caked signal and normalization fields, and let w_{ij} denote the selected-rod mask. The measured profile is

$$I_j^{\mathrm{rod}} = \frac{\sum_i w_{ij} S_{ij}}{\sum_i w_{ij} N_{ij}},$$

for bins with positive selected normalization. Signal and normalization are therefore summed before division. This prevents the final trace from becoming a sum of already normalized pixels and keeps the observable tied to the detector-space calibration.

The model profile is evaluated along the same selected branch. In compact notation,

$$M_j(\boldsymbol{\theta}_{\mathrm{dis}}) = B(x_j) + s\left[K * I_m^{\mathrm{nc}}(L; \boldsymbol{\theta}_{\mathrm{dis}})\right]_j,$$

where x_j is the profile coordinate, L is the corresponding stacking coordinate on the selected branch, B is a low-order background, s is a global scale, K is the fixed detector/profile broadening kernel, and I_m^{nc} is the no-correlation analytical stacking intensity from Sec. S4. When an explicit closed-form broadening kernel is used, K is written as a Gaussian-plus-Lorentzian kernel with widths fixed by the upstream resolution and mosaic/profile state or declared before the disorder fit. It is not used to release independent peak areas.

The disorder objective is evaluated only over the retained profile bins,

$$\chi^2_{\text{dis}}(\boldsymbol{\theta}_{\text{dis}}, s, B) = \sum_{j \in \mathcal{J}} \rho\left(\frac{I_j^{\text{rod}} - M_j(\boldsymbol{\theta}_{\text{dis}})}{\sigma_j}\right),$$

where ρ may be a least-squares or robust loss and $\mathcal{J}$ is the retained Q_z interval. The known Bragg locations on the selected branch are inherited from the fixed detector geometry and ordered model. Peak amplitudes are not fit independently before the stacking model is applied; otherwise the diffuse between-peak intensity would already have been absorbed by the preprocessing.

S3 Material background and stacking–fault model

In layered metal–halides such as PbI_2 each I–Pb–I sandwich behaves like a rigid trilayer that can shift laterally with negligible energy cost compared with the much stronger intra-layer bonding. Consequently the crystal can adopt several polytypes that differ only in the stacking sequence of these sandwiches along the c axis. Here we focus on the two sequences that bracket the experimentally relevant window:

S3.1 Polytypism of PbI_2

2H ($\boldsymbol{P}\bar{3}$m1) AB repeat stable below 140 °C; no lateral shift ($\phi_{AB} = 0$); see Fig. S1a.

6H ($\boldsymbol{R}\bar{3}$) ABCA repeat above ≈ 200 °C; lateral shift $a/\sqrt{3}$ gives $\phi = 2\pi/3$; 2H ⇔ 6H transition near 94 °C; see Fig. S1b.

Our samples never achieve perfect periodicity; stacks often contain finite 2H and 6H "domains" interspersed by stacking faults. To model the resulting diffuse scattering we adapt the Hendricks–Teller (HT) Markov formalism and follow the streamlined matrix notation introduced a few years later by Lifshitz, who showed how to re-express pair probabilities through the eigenvalues of a 3×3 transition matrix.

S3.2 Molecular structure factor for PbI_2

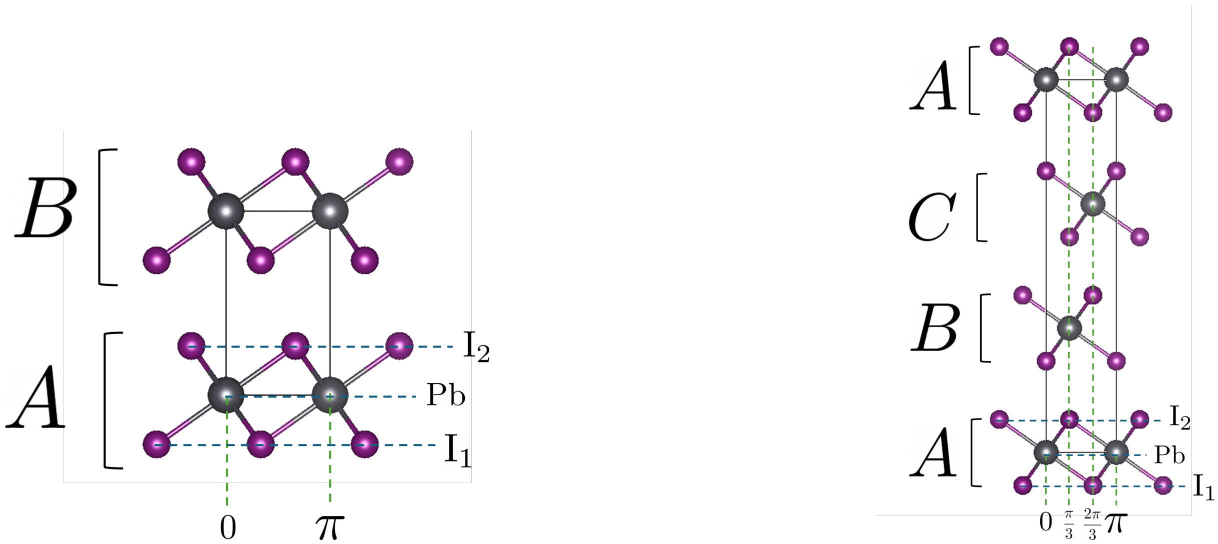
A single layer form factor is

$$F_m = b_{Pb} + b_I \left[e^{2\pi i(H/3+2K/3+Lz/3)} + e^{2\pi i(2H/3+K/3-Lz/3)}\right] \tag{S56}$$

Where now the amplitude will be

$$A = F_m \sum_{R_p} e^{i\vec{Q}\cdot\vec{R_p}} \tag{S57}$$

Where $\vec{R_p} = m\vec{a} + n\vec{b}$ is the in-plane position vector, with $m, n \in \mathbb{Z}$. Then $\vec{Q} \cdot \vec{R_p} \to Hm + kn$, and the Bragg condition makes H, K integers.



(a) 2H polytype (AB) with no in-plane phase shifts.

(b) 6H R$\bar{3}$ polytype (ABCA) with $\frac{1}{3}$-lattice translations giving $e^{i2\pi/3}$ phase steps.

Figure S1: Low-temperature 2H and high-temperature 6H stacking sequences in PbI_2. Arrows illustrate the lateral shifts responsible for the phase factors used in the HT–Lifshitz model.

S4 No-correlation stacking-fault model

Let $\tau_j \in \{0, 1\}$ indicate whether the j-th interface slips ($\tau_j = 1$) or not ($\tau_j = 0$). The structure factor of one layer is $F_0(h, k, \ell)$, and for a three-layer (ABC) repeat the per-layer phase advance is

$$\phi(h, k, \ell) = \frac{h + 2k}{3} + \ell/3, \tag{S58}$$

with h, k, ℓ in reciprocal-lattice units.
We now add the phase of each layer to the form factor from equation S57.

$$A = F_m \sum_{R_p} e^{i\vec{Q}\cdot\vec{R_p}} \sum_{\ell} e^{i\phi(h,k,\ell)\ell} \tag{S59}$$

Writing $e^{i\phi\ell} = \prod_{j=1}^{\ell} e^{i\phi}$

$$A = F_m \sum_{R_p} e^{i\vec{Q}\cdot\vec{R_p}} \prod_{\ell} e^{i\phi(h,k,\ell)\ell} \tag{S60}$$

S4.1 Coherent amplitude of an N-layer stack

At layer ℓ the factor multiplying F_0 is

$$\left[1 - \tau_\ell + \tau_\ell e^{-i\phi}\right] e^{i\phi\ell},$$

so that

$$A = F_0 \sum_{\ell=0}^{N-1} \left[\prod_{j=0}^{\ell} (1 - \tau_j + \tau_j e^{-i\phi})\right] e^{i\phi\ell}. \tag{S61}$$

S4.2 Uncorrelated faults

Assume the τ_j are i.i.d. with slip probability $p = \Pr(\tau_j = 1)$. Independence gives

$$\Big\langle \prod_{j=0}^{\ell} (1-\tau_j+\tau_j e^{-i\phi}) \Big\rangle = \big[(1-p)+p\,e^{-i\phi}\big]^{\ell} \equiv f^{\,\ell}, \qquad f \equiv (1-p)+p\,e^{-i\phi}. \tag{S62}$$

S4.3 Mean amplitude

Averaging (S61) and inserting (S62) gives the geometric series

$$\langle A\rangle = F_0 \sum_{\ell=0}^{N-1} f^{\ell} e^{i\phi\ell} = F_0 \frac{1-(f\,e^{i\phi})^N}{1-f\,e^{i\phi}}. \tag{S63}$$

S4.4 Infinite-crystal limit

For any non-zero fault probability ($|f|<1$),

$$\boxed{\lim_{N\to\infty} \langle A\rangle = F_0 \frac{f}{1-f\,e^{i\phi}}} \tag{S64}$$

The perfect-crystal result is recovered as $f \to 1$.

S4.5 Mean-square amplitude

The heights of sharp Bragg peaks: i.e. the square of the coherently averaged structure amplitude. This is only the part of the intensity that survives perfect phase coherence: sharp Bragg peaks. Any layer-to-layer fluctuations are averaged before squaring and thus disappear.

$$|\langle A\rangle|^2 = |F_0|^2 \frac{f^2}{|1-f\,e^{i\phi}|^2}$$

$$= |F_0|^2 \frac{f^2}{(1-f\cos\phi)^2+f^2\sin^2\phi} \tag{S65}$$

$$= |F_0|^2 \frac{f^2}{(1-f)^2+4f\sin^2\dfrac{\phi}{2}}. \tag{S66}$$

S4.6 Ensemble-averaged intensity

The full detector signal; adds the broadened peaks and diffuse background arising from stacking faults.

$$< |A|^2 > \tag{S67}$$

S4.6.1 Layer Autocorrelation

The layer-resolved autocorrelation of the stacking sequence tells you how phase coherence decays with layer phase shifts. It is the kernel that converts local disorder into the full intensity giving both Bragg and diffuse components.

Define

$$g_j \equiv 1 - \tau_j + \tau_j e^{-i\phi}, \qquad S_\ell \equiv \prod_{j=0}^{\ell} g_j. \tag{S68}$$

We wish to compute

$$\langle S_\ell S_{\ell'}^* \rangle = \langle \prod_{j=0}^{\ell} g_j \prod_{j'=0}^{\ell'} g_{j'}^* \rangle. \tag{S69}$$

Since the g_j are i.i.d., expectations of products factor, and $g_j g_j^* = 1$ identically. We split into two cases:

Case A: $\ell \geq \ell'$.

$$S_\ell S_{\ell'}^* = \underbrace{\Big(\prod_{j=0}^{\ell'} g_j\Big)\Big(\prod_{j'=0}^{\ell'} g_{j'}^*\Big)}_{=\ 1\ \text{(common layers)}} \underbrace{\Big(\prod_{j=\ell'+1}^{\ell} g_j\Big)}_{\text{extra layers}} \tag{S70}$$

- For $0 \leq j \leq \ell'$, $g_j g_j^* = 1$, so the first two products give 1. - For $j = \ell' + 1, \ldots, \ell$, independence implies

$$\Big\langle \prod_{j=\ell'+1}^{\ell} g_j \Big\rangle = \prod_{j=\ell'+1}^{\ell} \langle g_j \rangle = f^{\,\ell-\ell'}. \tag{S71}$$

Hence

$$\langle S_\ell S_{\ell'}^* \rangle = f^{\,\ell-\ell'}, \qquad (\ell \geq \ell'). \tag{S72}$$

Case B: $\ell < \ell'$. By an identical argument, swapping roles,

$$\langle S_\ell S_{\ell'}^* \rangle = (f^*)^{\ell'-\ell}, \qquad (\ell < \ell').$$

The Solutions are then

$$\boxed{\langle S_\ell S_{\ell'}^* \rangle = \begin{cases} f^{\,\ell-\ell'}, & \ell \geq \ell', \\ (f^*)^{\ell'-\ell}, & \ell < \ell'. \end{cases}} \tag{S73}$$

S4.7 Single-sum form of the double series

With $\ell'' = \ell - \ell'$, the double sum becomes

$$\sum_{\ell=0}^{N_z-1} \sum_{\ell'=0}^{N_z-1} f^{|\ell-\ell'|} e^{i(\phi-\psi)(\ell-\ell')} = N_z \sum_{\ell''=-N_z+1}^{N_z-1} f^{|\ell''|} e^{i(\phi-\psi)\ell''}. \tag{S74}$$

Splitting out $\ell'' = 0$ and the $\pm\ell''$ pairs yields

$$N_z \Bigg[1 + \sum_{\ell''=1}^{N_z-1} f^{\ell''} \big(e^{i(\phi-\psi)\ell''} + e^{-i(\phi-\psi)\ell''}\big)\Bigg],$$

with
$$\sum_{\ell''=1}^{N_z-1} f^{\ell''} e^{\pm i(\phi-\psi)\ell''} = \frac{f\, e^{\pm i(\phi-\psi)}\big[1 - f^{N_z-1} e^{\pm i(\phi-\psi)(N_z-1)}\big]}{1 - f\, e^{\pm i(\phi-\psi)}}.$$

S4.8 Infinite-crystal limit ($0 \le f < 1$)

Letting $N_z \to \infty$ gives

$$\sum_{\ell,\ell'} f^{|\ell-\ell'|} e^{i(\phi-\psi)(\ell-\ell')} = N_z \, \frac{1-f^2}{1+f^2-2f\cos(\phi-\psi)}. \tag{S75}$$

S4.9 Scattering cross-section

Substituting (S75),

$$\boxed{\frac{d\sigma}{d\Omega} = |F_m|^2 \, \frac{(2\pi)^2}{A_c} \, \delta(\mathbf{Q}-\mathbf{G}_p)\, N_p N_z \, \frac{1-f^2}{1+f^2-2f\cos(\phi-\psi)}}. \tag{S76}$$

This is identical to the Hendrix–Teller formula upon identifying $\phi^{\mathrm{HT}} = \phi - \psi$ and noting that their parameter $\cos(\phi^1 - \phi^2)$ equals f.